

List of Possible Proposals for M.Sc. Thesis

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Abstract

In the following document I'd share five proposals that I've spent time designing and thinking about. These proposals are not necessarily the best ones, but they are the ones that I think are interesting and feasible for an M.Sc. thesis.

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Chapter 1

Gravitational Waves: an Overview of the State

1.1 The Past

In 1915, Einstein presented the theory of general relativity. Incorporating his special relativistic view of the spacetime with the principle of equivalence (his happiest thought), he gave a geometrical description of spacetime, and gravitation as a curvature (in opposed to viewing it as a force between massive bodies), in which a freely falling body is not being pushed, but moves in the path of geodesics. Perhaps

“Spacetime tells matter how to move;
matter tells spacetime how to curve.”[49]

explains the essence of general relativity. The theory, despite its elegant mathematics and clean philosophy and principles, went on a slow path of earning its pride of place in physics. Though it explained the anomalous precession of Mercury’s perihelion; a first golden opportunity arose on 29 May 1919, when a total solar eclipse allowed the measurement of the deflection of light by the sun[24].

The observation were of poor quality, despite which, made Einstein, and his theory, a celebrity overnight. Later, on 22 September 1922, another set of observations, with outstanding quality, came out with precise agreement with General Relativity’s predictions[18]. From then on, additional tests has followed, first with the observation of gravitational redshift in astronomical measurements in 1954[57], and in laboratory in 1959[58], and later with the observation of gravitational time dilation in 1964[63].

However, the most spectacular prediction of general relativity, gravitational waves, remained elusive for decades.

In his 1916 paper, Einstein predicted the existence of gravitational waves[29], emerging from Einstein’s general theory of relativity. Just months after he formulated the field equations, in the linearized weak-field limit, he found wave-like solutions that propagate at the speed of light. However, the physical reality of these waves remained controversial for decades. Even Einstein himself wavered, publishing a paper in 1946 claiming gravitational waves didn’t exist.

The debate was finally settled in the 1950s through the work of Bondi, Pirani, and Robinson, who demonstrated that gravitational waves could carry energy and produce observable effects on test particles[15, 14, 35].

The theoretical understanding of gravitational radiation accelerated in the 1960s. The quadrupole formula, which describes the leading-order from slowly moving sources, provided the foundation for calculating energy loss from orbiting systems. Post-Newtonian approximations extended these calculations to higher orders, allowing predictions for how binary systems should evolve as they radiate energy. By the early 1970s, the stage was set for an indirect confirmation of gravitational waves through astrophysical observations.

1.1.1 The Dance of the Pulsars

The 1974 discovery of the binary pulsar PSR B1913+16 by Hulse and Taylor provided the first indirect evidence for gravitational radiation. This system—two neutron stars in a tight orbit—offered a natural laboratory for testing general relativity’s predictions[37]. As the system radiates gravitational waves, it should lose orbital energy, causing the orbital period to decrease at a precisely calculable rate. Over the following years, radio timing observations tracked this orbital decay with extraordinary precision[48]. By 1993, when Hulse and Taylor received the Nobel Prize, the observed decay matched general relativity’s predictions to better than 0.5%. Gravitational waves were real; they just hadn’t been directly detected[23].

1.1.2 Building the Ears

The quest for direct detection of gravitational waves began in earnest in the 1960s with Joseph Weber’s pioneering work on resonant bar detectors. Weber’s bars were massive aluminum cylinders designed to resonate in response to passing gravitational waves. Though Weber claimed detections in the late 1960s, his results were not reproducible, and the scientific community remained skeptical[67, 66].

The experimental work of Weber in 1969, together with his unverified claims of detection, spurred the development of more sensitive detectors. In the 1970s and 1980s, several groups around the world began constructing interferometric detectors, which promised orders of magnitude greater sensitivity than resonant bars. The Laser Interferometer Gravitational-Wave Observatory (LIGO) project, initiated in the late 1980s, aimed to build large-scale interferometers capable of detecting gravitational waves from astrophysical sources[68].

The idea lies behind the fact that given a passing gravitational radiation, spacetime is curved and therefore, lengths change. Using a pair of perpendicular (for getting the maximum length change) tunnels and two beams of in-phase photon lasers (so that the speed is constant); one can observe interference fringes produced when the light is com-

bined at the corner station shift back and forth as the wave changes in phase. This shift can be compared with the expectation from different types of signals to assess the probability that signal or noise is being observed[48].

The LIGO (Laser Interferometer Gravitational-Wave Observatory) project was proposed in the 1980s and approved by the U.S. Congress in 1992. Construction of two 4-kilometer facilities—one in Hanford, Washington, and one in Livingston, Louisiana—was completed in 1999. Initial LIGO operated from 2002 to 2010 without detecting gravitational waves, as expected; the technology wasn’t yet sensitive enough. The crucial upgrade to Advanced LIGO, incorporating decades of technical improvements, was completed in 2015.

Europe built the Virgo detector near Pisa, Italy, with 3-kilometer arms, which began operations in 2007 and underwent similar sensitivity upgrades. Japan constructed KAGRA, an underground 3-kilometer detector in the Kamioka mine, which joined the network in 2020. This global network of detectors is essential for localizing sources on the sky and testing the polarization properties of gravitational waves.

1.1.3 The New Astronomy

Four decades of thought, development, went on so in by the end of summer of 2015 the LIGO detectors were finally sensitive enough to detect plausible astrophysical sources. And on 14 September 2015 timing favored physicists again. Almost as soon as the detectors were turned on, they gave a signal that was strong enough to be an unmistakable source[2].

The signal—designated GW150914—swept through LIGO’s sensitive frequency band in a tenth of a second, matching theoretical predictions from numerical relativity simulations with remarkable fidelity. The waveform’s shape encoded the masses and spins of the black holes, the distance to the system, and even confirmed that the merged remnant obeyed general relativity’s predictions for black hole ringdown. This single detection accomplished multiple scientific goals

simultaneously: it confirmed gravitational waves exist, proved that binary black holes exist and merge within a Hubble time, and tested general relativity in the strong-field regime for the first time.

The announcement on 11 February 2016 marked the dawn of gravitational wave astronomy. The 2017 Nobel Prize in Physics was awarded to Rainer Weiss, Kip Thorne, and Barry Barish for their decisive contributions to the LIGO detector and the observation of gravitational waves.

But the story was just beginning. The truly transformative event came on August 17, 2017, with the detection of GW170817—gravitational waves from a binary neutron star merger[20]. Within seconds of the gravitational wave detection, the Fermi and INTEGRAL gamma-ray satellites detected a short gamma-ray burst from the same sky region. Within hours, optical telescopes located the host galaxy (NGC 4993) and observed the kilonova—the radioactive glow from newly synthesized heavy elements[31]. Over the following weeks, telescopes across the electromagnetic spectrum, from radio to X-rays, tracked the event’s evolution. This single event spawned over 100 papers in the first few months, addressing questions ranging from the neutron star equation of state to the cosmic origin of gold and platinum, from the Hubble constant to tests of fundamental physics.

GW170817 inaugurated the era of multimessenger astronomy with gravitational waves. It demonstrated that gravitational wave observations could guide electromagnetic follow-up, that joint observations could constrain astrophysics and cosmology in ways neither could alone, and that the gravitational wave community could coordinate with observers worldwide to maximize scientific return. The event also validated decades of theoretical work on neutron star mergers and showed that these systems are indeed factories for heavy element production.

1.2 The Present

The era of ground-based gravitational-wave detection with LIGO and Virgo is ongoing, with fur-

ther instrumental improvements already planned. One such upgrade is known as *A+* [21], which aims to improve broad band sensitivity through the use of quantum light squeezing. This technique reduces phase noise at high frequencies and radiation-pressure noise at low frequencies, thereby enhancing the detectors’ ability to observe a wider range of sources[48].

Gravitational-wave astronomy is not limited to the LIGO and Virgo observatories. Japan has constructed a 3-km underground interferometer, KAGRA, located in the Kamioka mine[1, 53, 10]. This detector represents the first implementation of cryogenic interferometric detection. KAGRA is intended to operate as part of the global gravitational-wave detector network alongside LIGO and Virgo[52]. In addition, the LIGO-India detector is expected to become operational in the mid-2020s. With detectors distributed globally, the network will achieve improved sensitivity to gravitational waves arriving from different directions and will operate closer to optimal performance[48].

Following GW170817 [4], the pace of discoveries accelerated. The first two observing runs (O1 and O2) detected 11 gravitational wave events [3]. The third observing run (O3), split into O3a and O3b from April 2019 to March 2020, detected approximately 80 additional events. The third Gravitational-Wave Transient Catalog (GWTC-3), released in 2021, included 90 confident detections [5]. These observations have revealed:

- **A diverse population of binary black holes**, ranging from stellar-mass systems to intermediate-mass mergers, with a wide range of mass ratios and spin configurations. The population properties provide constraints on formation channels, stellar evolution, and binary dynamics.
- **Binary neutron star systems** beyond GW170817**, though none with confirmed electromagnetic counterparts. These detections constrain the neutron star mass distribution and equation of state.

- **Neutron star-black hole binaries**, confirming that this channel contributes to the merger rate and enabling new tests of compact object physics.
- **Evidence for hierarchical mergers**—black holes that themselves resulted from previous mergers—suggesting dense stellar environments where repeated interactions can build up increasingly massive black holes.
- **Tests of general relativity** with increasing precision. So far, all observations are consistent with GR predictions, placing stringent constraints on alternative theories of gravity.

The field has matured to the point where multiple detections per week are routine during observing runs. The fourth observing run (*O4*) began in May 2023 with improved sensitivity across the network. The Japanese KAGRA detector joined at reduced sensitivity, working toward full integration. Plans for *O5*, potentially beginning in 2026–2027, promise further sensitivity improvements and higher detection rates.

1.3 Looking Ahead

Although predictions beyond 2025 are uncertain, plans for more advanced ground-based detectors are already under discussion. Two major proposals are the Einstein Telescope and the Cosmic Explorer. Both concepts require new facilities. The Einstein Telescope is planned as an underground triangular interferometer to mitigate seismic noise [30], while the Cosmic Explorer envisions a detector with arm lengths of approximately 40km, incorporating quantum squeezing techniques similar to those of *A+* and cryogenic cooling technologies developed for KAGRA and Voyager [33]. At present, funding for these projects has not been secured, and only early-phase studies have begun. Consequently, deployment is not expected before the 2030s.

In addition to the frequency range of approximately 10–2,000Hz probed by current and planned ground-based detectors, lower-frequency gravitational waves can be studied

indirectly through cosmic phenomena. These approaches provide access to gravitational-wave signals outside the reach of terrestrial interferometers.

1.3.1 Space-Based Detectors

To detect gravitational waves emitted by different classes of strong-gravity sources, space-based detectors are required. One such mission is the Laser Interferometer Space Antenna (LISA), a mission led by the European Space Agency with participation from NASA. LISA will consist of three spacecraft arranged in a triangular formation, orbiting the Sun while trailing the Earth and maintaining a precessing configuration.

LISA is expected to detect new classes of gravitational-wave sources, including mergers of black holes with masses in the range $10^4 M_\odot$ to $10^7 M_\odot$ located at galactic centers. The gravitational waves emitted by these systems contain information about the properties of black holes prior to and during galaxy mergers, contributing to astrophysical population studies.

Space-based gravitational-wave observations are particularly important due to the unique science they enable. Information about supermassive black holes during galaxy mergers can only be obtained through these observations. From a theoretical perspective, gravitational waves emitted in extreme-mass-ratio inspirals encode detailed information about the spacetime geometry of supermassive black holes.

Such measurements allow for tests of whether astrophysical black holes are described by the Kerr solution of the Einstein equations. Previous experience suggests that these observations are likely to confirm many predictions of general relativity, rule out certain modified gravity models, and constrain models of black hole formation. However, the possibility of unexpected results remains.

Historically, opening new observational windows has led to unforeseen discoveries. Gravitational-wave observations may provide insights into unresolved questions in gravitational physics, in-

cluding the late-time acceleration of the Universe, galactic rotation curves, and the unification of quantum mechanics with gravity.

The gravitational wave revolution is entering its second phase. The current detector network will continue to improve, eventually reaching design sensitivity and possibly being upgraded further. But the most dramatic developments will come from new facilities:

- **LISA (Laser Interferometer Space Antenna)**, formally adopted by ESA in January 2024, will launch in the mid-2030s to observe millihertz-frequency gravitational waves from space. LISA will detect supermassive black hole mergers, extreme mass ratio inspirals, and potentially stochastic backgrounds from the early universe—sources completely inaccessible to ground-based detectors due to seismic noise.
- **Einstein Telescope**, a proposed European underground facility with 10-kilometer arms, could improve sensitivity by a factor of 10 over current detectors [30]. Site selection is underway, with the Euregio Meuse-Rhine region (spanning Belgium, Netherlands, and Germany) and Sardinia, Italy, as leading candidates [32].
- **Cosmic Explorer**, a U.S. proposal for two facilities with 40-kilometer and 20-kilometer arms, would achieve similar sensitivity improvements [33]. The longer arms enable detection of binary black holes across cosmic time, potentially observing the first generation of black holes formed in the universe.

These next-generation facilities will not merely detect more events—they will enable qualitatively new science. With sensitivities 10 times better than current detectors, they will observe every stellar-mass binary black hole merger in the observable universe, detect neutron star mergers out to cosmological distances, precisely test general relativity in regimes currently inaccessible, and potentially discover unexpected sources. The computational challenges of handling tens

of thousands of detections per year, extracting maximum information from each observation, and testing theoretical predictions with unprecedented precision will define much of the research landscape for the coming decades.

Additionally, pulsar timing arrays have recently reported evidence for a stochastic gravitational wave background at nanohertz frequencies [8, 28, 65], likely from supermassive black hole binaries across the universe. This opens yet another frequency window for gravitational wave observations, complementing ground-based and space-based detectors.

The gravitational wave revolution has moved from first detection to regular observations to planning for transformative new capabilities—all in less than a decade. The field is simultaneously harvesting the scientific returns from current detections and preparing for facilities that will observe the gravitational wave sky in exquisite detail. For researchers entering the field now, this creates a unique opportunity: contributing to both near-term challenges with current data and laying groundwork for the next generation of discoveries.

Chapter 2

SBI Pipeline for Constraining Cosmic String Microphysics from NANOGrav Pulsar Timing Array Data

2.1 Abstract

The nano-hertz gravitational wave background recently reported by NANOGrav [7] and other pulsar timing arrays (PTAs) collaborations stands as one of the most exciting signals in contemporary physics. Among the cosmological interpretations, a stochastic gravitational wave background (SGWB) produced by a network of cosmic strings (one-dimensional topological defects formed during symmetry-breaking phase transitions in the early universe [40]) is a well-motivated and observationally relevant candidate. The microphysics of such a network, encoded in the string tension $G\mu$, the reconnection probability p and the loop distribution model, leaves a characteristic imprint on the spectral energy density $\Omega_{\text{gw}}(f)$.

Traditional Bayesian parameter estimation using ENTERPRICE/PTArcade relies on MCM sampling with an explicit likelihood [27, 50], which scales poorly as the model complexity grows and struggles to marginalize efficiently over nuisance parameters such as the competing supermassive black hole binary (SMBHB) background.

In this proposal, we develop a fully open-source, modular **Simulation-Based Inference (SBI)** pipeline [22] targeting the PTA frequency band. The forward simulator encodes the Velocity-dependent One-Scale (VOS) network evolution equations [44] in an expanding FLRW background, computes the resulting SGWB spectral density analytically (via the BOS [13] and LRS [42]

loop distribution models), and injects the signal into mock NANOGrav-15ye noise [39]. The inference backend employs Truncated Marginal Neural Ratio Estimation (TMNRE) [47], implemented via the swyft library [46], to learn the marginal posteriors over the string microphysics parameters.

This approach is explicitly complementary to the existing saqqara library [11], which targets LISA and generic signal templates; our contribution is a dedicated PTA-band forward model and a cosmic-string-specific signal module that could ultimately be contributed upstream.

The cluster resources available to the group are utilized for the large-scale parallel simulation campaigns needed to train the neural estimator.

2.2 Objectives

- Implement a vectorized, physically complete forward simulator for the SGWB from cosmic string networks in the PTA frequency band, based on the VOS model [43, 44] and Nambu-Goto loop dynamics;
- Develop a modular SBI pipeline (forward simulator + noise model + TMNRE inference) as a reusable open-source software package, designed to be architecturally compatible with the saqqara library [11] for future integration;

- Constrain cosmic string microphysics parameters (G_μ , p , loop distribution model) using the latest NANOGrav-15yr dataset [7];
- Benchmark the SBI pipeline against traditional MCMC-based ENTERPRISE/PTArcade results [27, 50] to validate and quantify the improvement in inference efficiency;
- Explore the implicit marginalization capability of TMNRE [47] over the SMBHB contamination as a nuisance source.
- Posterior predictive checks and simulation-based calibration (SBC) [64] are applied to validate inference quality.
- Results are compared to the existing PTArcade/ENTERPRISE analyses of the same dataset [7, 50] for quantitative benchmarking.

2.3 Methods

- The VOS model [44] (a pair of coupled ODEs describing the correlation length L and RMS velocity v of the string network in an FLRW background) is solved numerically using a stiff ODE integrator. This constitutes the physically relativistic core of the forward mode;
- The SGWB spectral energy density $\Omega_{\text{gw}}(f; G_\mu, p)$ is computed for integrating the loop number density (BOS [13], LRS [42], or Lorenz-Wands-Martins [61] models) weighted by the GW emission power from cusps and kinks.
- The resulting signal is injected into noise model based on the NANOGrav 15-year free-spectrum posterior [39], using the enterprise [27] + enterprise_extensions PTA likelihood frameworks as a reference.
- A compression network (1D CNN or UNet architecture, following the saqqara design [11]) is trained to map frequency-domain PTA data to a low-dimensional summary embedding. A neural ratio estimator is then trained using swyft [46] to produce marginal posteriors on the parameter vector $\theta = \{\log G_\mu, \log p, \text{loop model index}\}$.
- Simulation campaigns are distributed across the group’s compute cluster for embarrassingly parallel prior sampling.

Chapter 3

Beyond the Power Spectrum: Morphological Summary Statistics for Stochastic Gravitational Wave Background Inference in Pulsar Timing Arrays

3.1 Abstract

Parameter estimation from pulsar timing array data is predominantly performed using the cross-power spectral density as the summary statistic fed into a likelihood or inference algorithm [60]. While the power spectrum captures the mean spectral shape, it discards higher-order information encoded in the non-Gaussian features of the signal. In cosmological large-scale structure analysis, morphological measures — specifically Minkowski functionals [45], persistent homology [25], and weighted morphological statistics [?] — have been shown to carry significant additional constraining power beyond the power spectrum, particularly in the non-Gaussian regime [38, 70].

This proposal investigates the application of these topological and geometrical summary statistics to SGWB data analysis in the context of PTA experiments. The key observation is that the SGWB, as reconstructed from PTA cross-correlations across frequency bins and pulsar pairs, forms a two-dimensional map (pulsar pairs $\times$ frequency) whose morphological structure encodes information about the signal’s spectral shape, angular distribution, and non-Gaussianity. We propose to implement a systematic comparison of standard (power spectrum) and morphological summary statistics as inputs to a Simulation-Based Inference framework [22], quantifying via Fisher fore-

casts and SBI posterior widths which combination of statistics yields the tightest constraints on cosmological source parameters.

The target sources include the cosmic string network and a simple power-law SGWB as a baseline. This work establishes a direct bridge between Prof. Movahed’s morphological statistics toolkit [38, 6] and gravitational wave data analysis, and is designed to yield a methods paper regardless of the physical source chosen.

3.2 Objectives

- Implement Minkowski functionals [45] and persistent homology [25] as summary statistics operating on the frequency–pulsar-pair cross-correlation matrix of PTA residuals;
- Quantify via Fisher information matrix analysis the information gain of morphological statistics relative to the power spectrum alone for cosmic string parameter estimation [70];
- Build a full SBI pipeline [22, 11] that uses these combined statistics as the compressed data representation fed to a neural ratio estimator;
- Identify the optimal combination of summary statistics as a function of signal-to-noise ratio

and source model, producing a general result applicable to any SGWB source [60].

3.3 Methods

- Mock PTA datasets are generated by injecting SGWB signals (cosmic string + SMBHB) into NANOGrav-15yr-calibrated noise realizations [7, 39], for a wide prior over source parameters.
- For each realization, the frequency-domain cross-correlation matrix $C(f, a, b)$ (pulsars a, b) is computed. This matrix is the primary data object.
- Minkowski functionals (area, perimeter, Euler characteristic as a function of threshold) [45] are computed on excursion sets of $C(f)$. Persistent homology [25] is computed using the Vietoris-Rips filtration of the frequency-pulsar graph.
- A Fisher information matrix forecast is constructed semi-analytically using the Gram matrix of summary statistic derivatives with respect to the source parameters [70, 38].
- An SBI pipeline (TMNRE via `swyft` [47, 46]) is constructed using each summary statistic set in turn, and posterior volumes are compared as a direct measurement of information content [11].
- The cluster is used for the simulation campaign ($\mathcal{O}(10^4\text{--}10^5)$ simulations) and for parallelized morphological statistic computation.

Chapter 4

PTAforge: A Modular, High-Performance Forward Simulation Framework for New-Physics SGWB Signals in Pulsar Timing Arrays

4.1 Abstract

The growing maturity of pulsar timing array experiments — NANOGrav [7], EPTA [12], PPTA [59], CPTA [69], and the emerging IPTA [55] — demands analysis software that is physically complete, computationally efficient, and extensible to novel signal models from beyond-standard-model physics. The existing tool PTArcade [51] provides a user-friendly wrapper around ENTERPRISE [27] for MCMC-based new-physics searches. However, no general-purpose forward simulation library exists that is purpose-built for generating large-scale simulation campaigns from diverse cosmological SGWB source models, which are the prerequisite for Simulation-Based Inference workflows [22].

This proposal develops **PTAforge**: an open-source, modular, high-performance Python/C++ simulation framework for PTA SGWB signals. PTAforge separates the physics layer (signal model: cosmic strings, phase transitions, primordial GWs, etc.), the detector layer (PTA noise, pulsar timing residuals, antenna response), and the data product layer (cross-power spectral density, Hellings-Downs correlations [34], time-domain residuals) into independently testable, composable modules. The framework is designed to serve as the simulation backend for SBI workflows (compatible with saqqara [11] and swyft [46]) and as a standalone signal injection tool for traditional ENTER-

PRISE [27]/PTArcade [51] analyses.

The contribution here is explicitly a software contribution: the physics is well-established, but the infrastructure to run it at scale, with reproducible outputs, modular signal composition, and GPU-accelerated batch generation, does not exist. This addresses a real gap in the PTA software ecosystem and is publishable as a methods/software paper independent of any specific physical result.

4.2 Objectives

- Design and implement the PTAforge architecture: signal module interface, detector module interface, and data product module, with clean separation of concerns and a plugin system for new signal models;
- Implement a library of built-in signal models: power-law SGWB [60], cosmic string (VOS/-BOS/LRS) [44, 13, 42], first-order phase transition [19], and primordial inflation as baseline plugins;
- Achieve GPU-accelerated batch simulation via JAX [16] or PyTorch backends, targeting $\mathcal{O}(10^4)$ simulations per minute on a single GPU node, enabling practical SBI [22] simulation campaigns;

- Validate PTAforge outputs quantitatively against PTArcade [51]/ENTERPRISE [27] results on shared benchmarks using the NANOGrav 15yr dataset [7, 39];
- Release as a fully documented open-source package with continuous integration, contributing signal modules back to the saqqara [11] ecosystem.

4.3 Methods

- The software architecture is designed following the modular pattern established by saqqara [11]: a `SignalModel` abstract base class, a `NoiseModel` abstract base class, and a `Simulator` class that composes them and produces batched outputs.
- All computationally intensive inner loops (loop number density integrals, power spectrum evaluations, PTA response functions) are implemented in JAX [16] for JIT compilation and automatic batching via `vmap`.
- A Python plugin registry allows user-defined signal models to be registered and used seamlessly within the SBI [22] or MCMC workflow without modifying core library code.
- Unit tests, integration tests, and numerical benchmarks are implemented via `pytest` and run in CI. Outputs for the power-law SGWB case are compared against PTArcade’s [51] known posteriors to validate correctness.
- Documentation is built with Sphinx and includes worked examples for both SBI (swyft [46] integration) and MCMC (PTArcade [51]/ENTERPRISE [27] integration) workflows.
- The cluster is used for performance benchmarking of the GPU batch generation and for running the validation simulation campaigns.

Chapter 5

Simultaneous Source Separation and Parameter Estimation for Overlapping SGWB Signals in PTA Data via Simulation-Based Inference

5.1 Abstract

The gravitational wave background detected by pulsar timing arrays is almost certainly a superposition of multiple sources: at minimum, an astrophysical background from supermassive black hole binaries (SMBHBs) [62], and potentially one or more cosmological backgrounds from phase transitions [19], cosmic strings [26], or primordial inflation. Disentangling these overlapping signals is one of the central challenges in PTA data analysis [7]. Traditional approaches either fix the astrophysical background and search for a cosmological excess, or perform joint MCMC sampling over the combined parameter space [51, 27] — the latter becoming computationally intractable as the number of source classes grows.

Simulation-Based Inference offers a principled solution [22]: by simulating overlapping signal realizations from the joint prior over all source parameters and training a neural ratio estimator [47] to learn the marginal posteriors, one can achieve implicit marginalization over the contaminating source without any explicit likelihood assumption. This proposal applies this strategy to the specific problem of cosmic string signal recovery in the presence of SMBHB contamination in NANOGrav data, extending the proof-of-concept demonstrated for LISA transients in Alvey et al. (2024) [11] to the PTA band and to a physically motivated cosmological source.

This is the most directly scientifically impactful of the proposed theses, as it addresses a real open problem in PTA data analysis and produces constraints on $G\mu$ that are robust to SMBHB model uncertainty.

5.2 Objectives

- Model the joint SGWB from cosmic strings [44, 13] and SMBHBs [62] in the PTA frequency band, parametrizing both source classes;
- Demonstrate that TMNRE [47] with implicit marginalization can recover the marginal posterior on cosmic string parameters while treating SMBHB parameters as nuisance variables [22];
- Apply the pipeline to mock NANOGrav 15yr data [7, 39] and compare constraints on $G\mu$ with and without explicit SMBHB marginalization;
- Quantify the bias introduced in cosmic string parameter estimates when the SMBHB background is incorrectly modeled or ignored [9];
- Optionally apply to the real NANOGrav 15yr free-spectrum chains as a data compression/re-analysis step [7].

5.3 Methods

- The forward simulator generates joint SGWB realizations by drawing simultaneously from the cosmic string prior ($\log G\mu$, loop model) [44, 13, 42] and the SMBHB prior (amplitude A , spectral index γ) [62] and summing the two spectral contributions before injecting into PTA noise [39].
- The TMNRE algorithm in `swyft` [47, 46] is used with the SMBHB parameters treated as implicit nuisance dimensions — the neural ratio estimator learns the marginal likelihood ratio $p(\mathbf{x}|\theta_{\text{CS}})/p(\mathbf{x})$ where $\mathbf{x}$ is the data and θ_{CS} are only the cosmic string parameters.
- This is validated against an explicit joint posterior (obtained by sampling over all parameters) to verify that the marginals agree, and the number of required simulations is compared [11].
- A second experiment injects only a cosmic string signal but fits with a model that assumes SMBHB contamination, to quantify the bias from incorrect noise modeling [9] — a practically important diagnostic.
- Fisher forecasts are computed for the joint model to understand which parameter combinations are fundamentally degenerate [62].
- The cluster enables the large simulation campaigns required ($\sim 10^5$ samples from the joint prior) [22].

Chapter 6

Differentiable Gravitational Wave Cosmology: A JAX-Based Framework for Gradient-Informed Inference of SGWB Source Parameters

6.1 Abstract

Modern machine learning toolchains built on differentiable programming — PyTorch, JAX [16] — have transformed computational physics by enabling gradient-based optimization over physical simulators. In gravitational wave cosmology, the forward model mapping source parameters to the observed SGWB spectral density is a smooth, differentiable function of those parameters. Yet existing PTA analysis frameworks (ENTERPRISE [27], PTArcade [51]) treat this model as a black box, using gradient-free samplers (MCMC) and discarding the information in the Jacobian of the forward model. Simulation-Based Inference methods [22], while powerful, also do not exploit the differentiability of the simulator.

This proposal develops a JAX-based [16] differentiable forward simulator for SGWB signals in the PTA band, and investigates three inference strategies that exploit gradients: (1) gradient-based MCMC (NUTS/HMC [36]) via NumPyro [56]; (2) variational inference (VI) using normalizing flows [54] with gradient-optimized parameters; and (3) Neural Posterior Estimation (NPE) with gradient-informed sequential simulation (active learning) [22]. The result is both a software contribution (a publicly available differentiable SGWB simulator) and a methodological comparison — identifying which inference strategy is most simulation-efficient for cosmic string parameter

estimation under realistic PTA noise [7].

This proposal is the most technically ambitious and software-forward of the five. It is appropriate if the student’s primary interest is in the architecture of inference systems and differentiable scientific computing, with the GW physics as the application domain.

6.2 Objectives

- Implement a fully differentiable SGWB forward model for cosmic strings [44, 13] and a power-law SGWB in JAX [16], supporting `jit`, `grad`, `vmap`, and `pmap` transformations;
- Deploy gradient-based MCMC (HMC/NUTS [36] via NumPyro [56]) on the differentiable model and benchmark convergence against gradient-free MCMC (PTMCMCSampler [27]);
- Implement a normalizing flow posterior [54] (using `nflows` or `flowjax`) trained via maximum likelihood on simulated (θ, x) pairs, and compare to TMNRE [47];
- Implement a gradient-informed active learning loop for sequential simulation [22], reducing the number of simulator calls needed to achieve target posterior accuracy;

- Release the differentiable simulator as a standalone JAX [16] library compatible with the broader NumPyro [56]/BlackJAX [17] inference ecosystem.

6.3 Methods

- The VOS network evolution equations [44] and the $\Omega_{\text{gw}}(f)$ integral are implemented in JAX [16] as pure functions, verified to be differentiable through `jax.grad` by comparison with finite-difference derivatives.
- Automatic differentiation through the ODE solver is achieved via the `difffrax` library [41] (Kidger 2021), which provides differentiable Dormand-Prince and Kvaerno solvers in JAX.
- For gradient-based MCMC, the PTA likelihood (Gaussian approximation to the free-spectrum posterior [39]) is implemented in NumPyro [56] and NUTS [36] is run using BlackJAX [17], with wall-clock time and effective sample size compared to PTMCMCSampler.
- For normalizing flows [54], a masked autoregressive flow is trained offline on $(\theta, \text{summary}(x))$ pairs, then used as an amortized posterior. Posterior quality is assessed via simulation-based calibration [64].
- The gradient-informed active learning loop selects new simulation points from regions of high posterior uncertainty (estimated via an ensemble of normalizing flows [54]), reducing simulation budget by targeted prior refinement [22].
- All experiments are run on the group’s GPU cluster nodes; the JAX [16] implementation enables transparent multi-GPU distribution via `pmap`.

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